Master Program in *Data Science and Business Informatics*

Statistics for Data Science

Project Presentation – ALE

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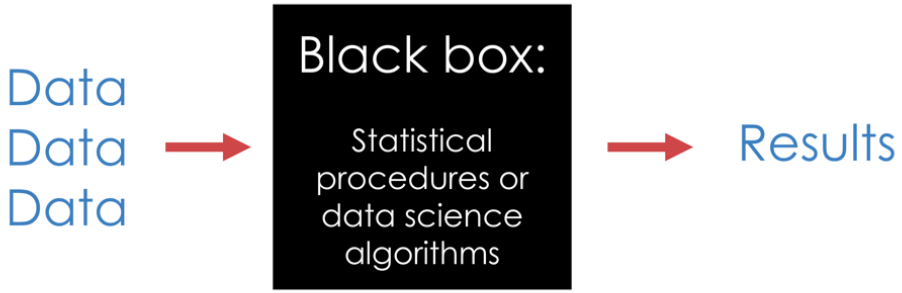
Black Box Supervised Learning Models



- **E.g.:** trees, random forests, nearest neighbors, neural networks etc…
- **Problem:** difficult to interpret and to understand the effects of the predictors on the predicted response.
- Suppose we have fit a supervised learning model for approximating $E[Y|\boldsymbol{X} = \mathrm{x}] \approx f(\mathrm{x})$, with:
  - Y: scalar response;
  - $\boldsymbol{X} = (X_1, X_2, \dots, X_d)$: vector of *d* predictors;
  - $f(\cdot)$: fitted model that predicts Y as a function of $\boldsymbol{X}$
- **Objective:** to visualize and understand the main effects dependence of $f(\mathbf{x}) = f(x_1, x_2, \dots, x_d)$ on each of the individual predictors, and the low-order interaction effects among pairs of predictors.

Partial Dependence Plots

PD Plots (d = 2)

To understand the effect of X_1 on the predicted response, a PD plot is a plot of the function

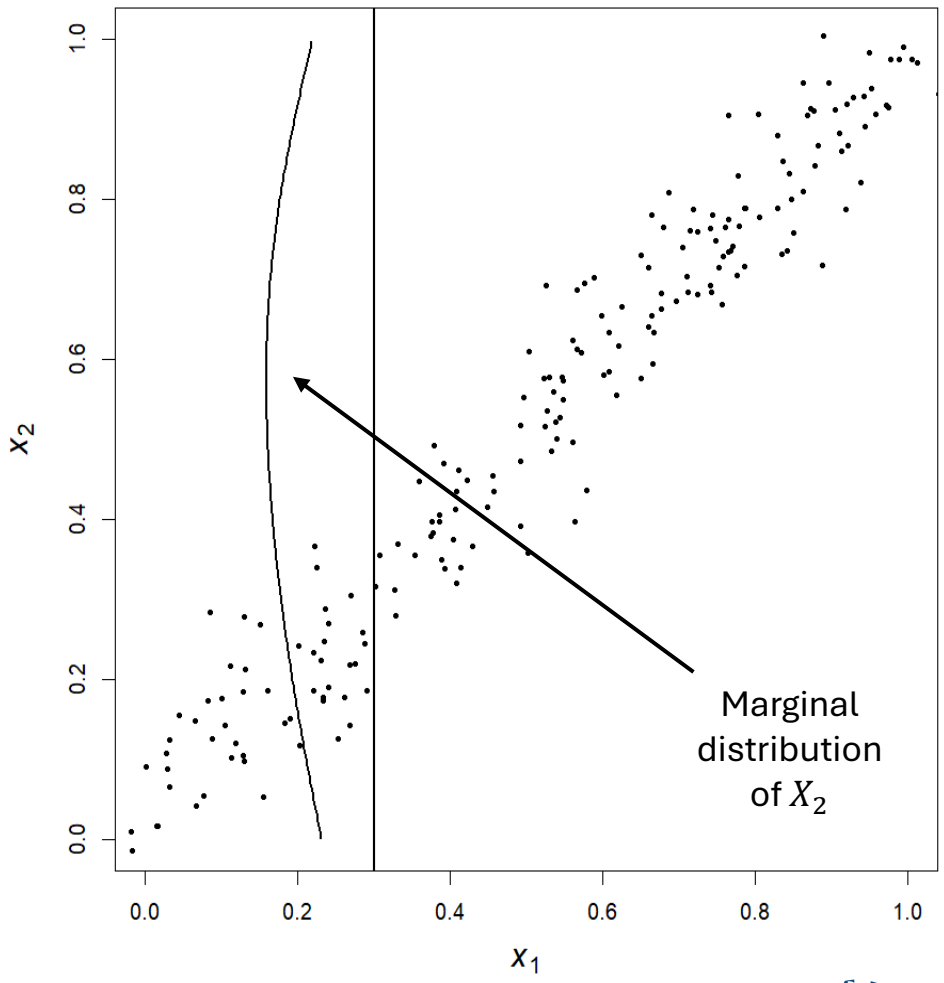
$$f_{1,PD}(x_1) \equiv \mathrm{E}[f(x_1, X_2)] = \int p_2(x_2)\, f(x_1, x_2)\, dx_2$$

versus x_1, where $p_2(\cdot)$ denotes the marginal distribution of X_2.

- An estimate of this formula is:

$$\hat{f}_{1,PD}(x_1) \equiv \frac{1}{n}\sum_{i=1}^{n} f\left(x_1, x_{i,2}\right).$$

- The integral is the weighted average of $f(x_1, X_2)$ as X_2 varies over its marginal distribution.
- **Problem:** it requires rather severe extrapolation beyond the envelope of the training data in the presence of *multicollinearity*;
  - ➢ PD pot unreliable indicator of the effect of x_1.



Marginal Plots

M Plots ($d = 2$)

An M plot of the effect of X_1 is a plot of the function

$$f_{1,M}(x_1) \equiv \mathrm{E}[f(X_1, X_2)|X_1 = x_1] = \int p_{2|1}(x_2|x_1)\, f(x_1, x_2)\, dx_2$$

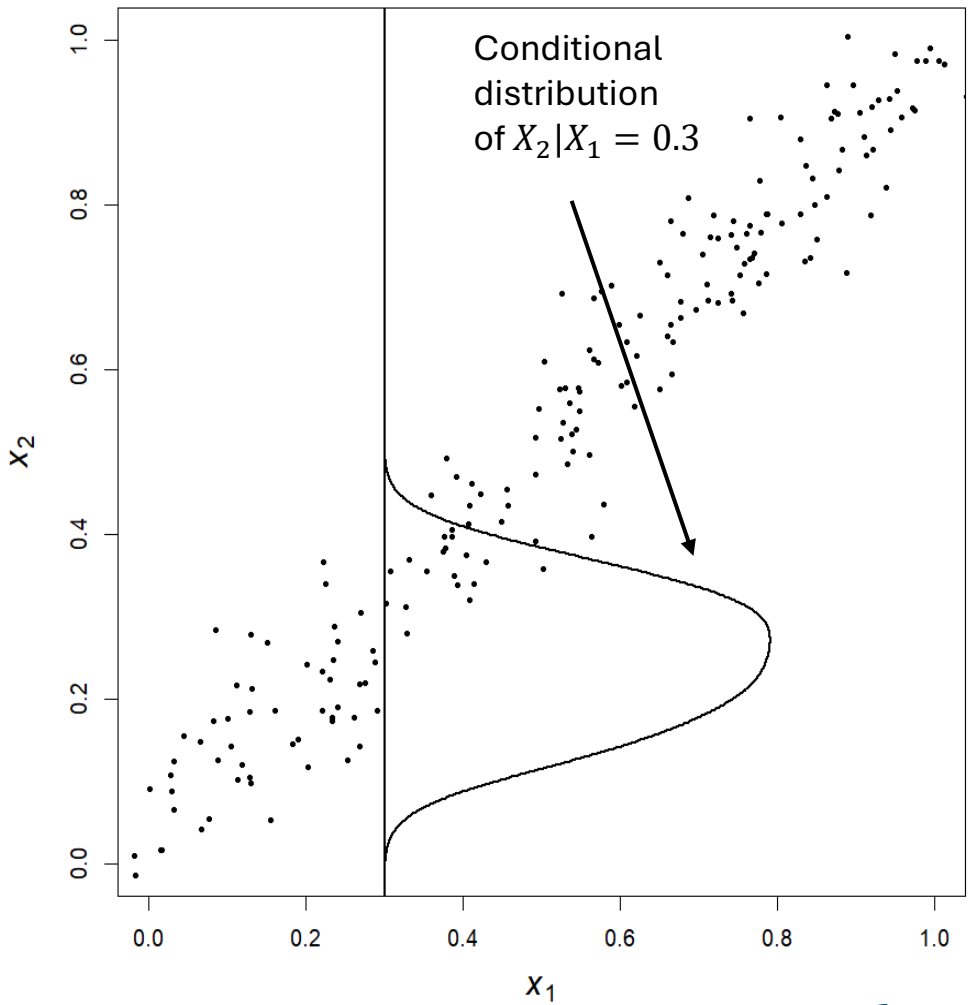
versus x_1, where $p_{2|1}(\cdot\,|\,\cdot)$ denotes the conditional distribution of X_2 given X_1.

- An estimate of $f_{1,M}(x_1)$ is:

$$\hat{f}_{1,M}(x_1) \equiv \frac{1}{n(x_1)} \sum_{i \in N(x_1)} f(x_1, x_{i,2})$$

  - $N(x_1)$ subset of row indices *i* for which $x_{i,1}$ falls into some small, appropriately selected neighborhood of x_1;
  - $n(x_1)$ number of observations in the neighborhood.

- They avoid the extrapolation problem of the PD plots.
- **Problem:** $f_{1,M}(x_1)$ will reflect the effect of both x_1 and x_2 as a consequence of the *omitted variable bias* phenomenon in regression.



ALE Plots

Accumulated Local Effect (ALE)

The ALE main-effect of X_1, for the case that *d*=2 and $f(\cdot)$ is differentiable, is defined as

$$f_{1,ALE}(x_1) \equiv \int_{x_{min,1}}^{x_1} \mathrm{E}[f^1(X_1, X_2)|X_1 = z_1]\, dz_1 - c = \int_{x_{min,1}}^{x_1} \int p_{2|1}(x_2|z_1) f^1(z_1, x_2) dx_2 dz_2 - c$$

where $f^1(x_1, x_2) \equiv \frac{\partial f(x_1,x_2)}{\partial x_1}$ represents the *local effect* of x_1 on $f(\cdot)$ at (x_1, x_2) and $x_{min,1}$ is some value chosen near the lower bound of the effective support of of $p_1(\cdot)$.

- The use of the conditional density $p_{2|1}(x_2|x_1)$, instead of the marginal density $p_2(x_2)$, avoids the extrapolation required in PD plots.
  - Avoidance of extrapolation similar to M plots.
- By averaging (across x_2) and accumulating (up to x_1) the local effects, as opposed to directly averaging $f(\cdot)$, it avoids the omitted nuisance variable bias of the M plots.

ALE Main Effect

Uncentered ALE Main Effect

Consider any j $\in \{1,2,\dots,d\}$, and suppose the sequence of partitions $\{P_j^K : K = 1,2,\dots\}$ is such that $\lim_{K\to\infty} \delta_{j,K} = 0$. When $f(\cdot)$ and p are such that the following limit exists and is independent of the particular sequence of partitions, the uncentered ALE main (first-order) effect function of X_j is defined as (for $x_j \in S_j$)

$$g_{j,ALE}(x_j) \equiv \lim_{K\to\infty} \sum_{k=1}^{k_j^K(x_j)} \mathrm{E}\left[f\left(z_{k,j}^K, \boldsymbol{X}_{\backslash \mathrm{j}}\right) - f\left(z_{k-1,j}^K, \boldsymbol{X}_{\backslash \mathrm{j}}\right) | X_j \in \left]z_{k-1,j}^K, z_{k,j}^K\right]\right].$$

- *Theorem*. ***Uncentered ALE Main Effect for differentiable $f(\cdot)$***
  - *i.* $f\left(x_j, \boldsymbol{x}_{\backslash \mathrm{j}}\right)$ is differentiable in x_j on S (compact support);
  - *ii.* $f^j\left(x_j, \boldsymbol{x}_{\backslash \mathrm{j}}\right)$ is continuous in $\left(x_j, \boldsymbol{x}_{\backslash \mathrm{j}}\right)$ on S;
  - iii. $\mathrm{E}\left[f^j\left(X_j, \mathbf{X}_{\backslash \mathrm{j}}\right) | X_j = z_j\right]$ is continuous in z_j on S_j.

  Then, for each $x_j \in S_j$,

$$g_{j,ALE}(x_j) = \int_{x_{min,j}}^{x_j} \mathrm{E}\left[f^j\left(X_j, \mathbf{X}_{\backslash \mathrm{j}}\right) | X_j = z_j\right] dz_j.$$

- **Centered:** center so that $f_{j,ALE}(X_j)$ has a mean of zero w.r.t. the marginal distribution of X_j:

$$f_{j,ALE}(x_j) \equiv g_{j,ALE}(x_j) - \int p_j(z_j) g_{j,ALE}(z_j) dz_j$$

ALE Second-Order Effect

Uncentered ALE Second-Order Effect

Consider any pair $\{j,l\} \subseteq \{1,2,\dots,d\}$, and corresponding sequences of partitions $\{P_j^K : K = 1,2,\dots\}$ and $\{P_l^K : K = 1,2,\dots\}$ such that $\lim_{K\to\infty} \delta_{j,K} = \lim_{K\to\infty} \delta_{l,K} = 0$. When $f(\cdot)$ and p are such that the following limit exists and is independent of the particular sequences of partitions, the uncentered ALE second-order effect function of (X_j, X_l) is defined as (for $(x_j, x_l) \in S_j \times S_l$).

$$h_{\{j,l\},ALE}(x_j, x_l) \equiv \lim_{K\to\infty} \sum_{k=1}^{k_j^K(x_j)} \sum_{m=1}^{k_l^K(x_l)} \mathrm{E}\left[\Delta_f^{\{j,l\}}(K,k,m;\boldsymbol{X}_{\setminus\{j,l\}}) | X_j \in \left]z_{k-1,j}^K, z_{k,j}^K\right], X_l \in \left]z_{m-1,j}^K, z_{m,j}^K\right]\right].$$

- *Theorem*. ***Uncentered ALE Second-Order effect for differentiable $f(\cdot)$***
  - Analogous to Uncentered ALE Main Effect for differentiable $f(\cdot)$ theorem.

    For each $(x_j, x_l) \in S_j \times S_l$

$$h_{\{j,l\},ALE}(x_j, x_l) = \int_{x_{min,l}}^{x_l} \int_{x_{min,j}}^{x_j} \mathrm{E}\left[f^{\{j,l\}}(X_j, X_l, \boldsymbol{X}_{\setminus\{j,l\}}) | X_j = z_j, X_l = z_l\right] dz_j dz_l \,.$$

- **Centered:** doubly center $h_{\{j,l\},ALE}(x_j, x_l)$ subtracting both of its first-order effects of X_j and X_l, and then, for this resulting function, subtracting its zero-order effect.
  - The ALE higher-order effects are defined analogously

Estimation

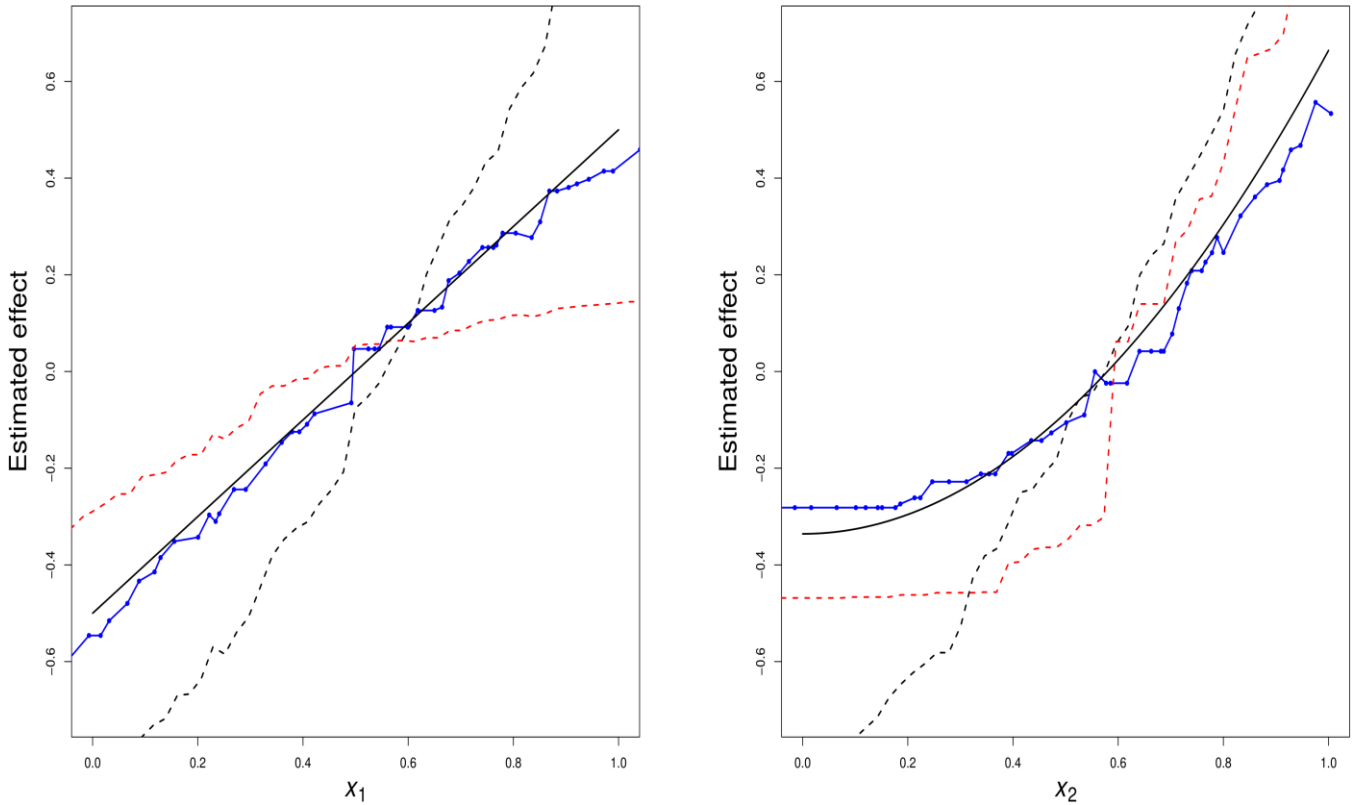
- Estimation of $f_{j,ALE}(x_j)$ and $f_{\{j,l\},ALE}(x_j, x_l)$:
  i. replace the sequence of partition by some appropriate fixed partition of the sample range $\{\mathrm{x}_{i,J} : i = 1,2, \dots, n\}$;
  ii. replace the conditional expectations by sample averages across $\{\mathrm{x}_{i,\setminus J} : i = 1,2, \dots, n\}$, conditioned on $\mathrm{x}_{i,J}$ falling into the corresponding interval/cell of the partition.
- For each j ∈ $\{1,2, \dots, d\}$, let $\{N_j(k) =]z_{k-1,j}, z_{k,j}] : k = 1,2, \dots, K\}$ be a sufficiently fine partition of the sample range of $x_{i,j}$ into *K* intervals.
  - **E.g.**: $z_{k,j}$ can be chosen as the $\frac{k}{K}$ quantile of the empirical distribution of $x_{i,j}$.
- For general *d*, an estimate of the uncentered ALE main effect is:

$$\hat{g}_{j,ALE}(x) = \sum_{k=1}^{k_j(x)} \frac{1}{n_j(k)} \sum_{\{i: x_{i,j} \in N_j(k)\}} \left[f\left(z_{k,j}, \mathrm{x}_{\mathrm{i},\setminus \mathrm{j}}\right) - f\left(z_{k-1,j}, \mathrm{x}_{i,\setminus \mathrm{j}}\right) \right]$$

- Then center by subtracting an estimate of $\mathrm{E}[g_{j,ALE}(X_j)]$ $\left(= \frac{1}{n} \sum_{k=1}^{K} n_j(k) \hat{g}_{j,ALE}(z_{k,j})\right)$.
- To estimate the ALE second-order effect of a pair of predictors (X_j, X_l), the sample range of $(x_{i,j}, x_{i,l})$ is partitioned into a grid of K^2 rectangular cells obtained as the Cartesian product of the individual 1D partitions. Then, a procedure analogous to case 1D is followed.

Experiment 1 – Tree model with synthetic data

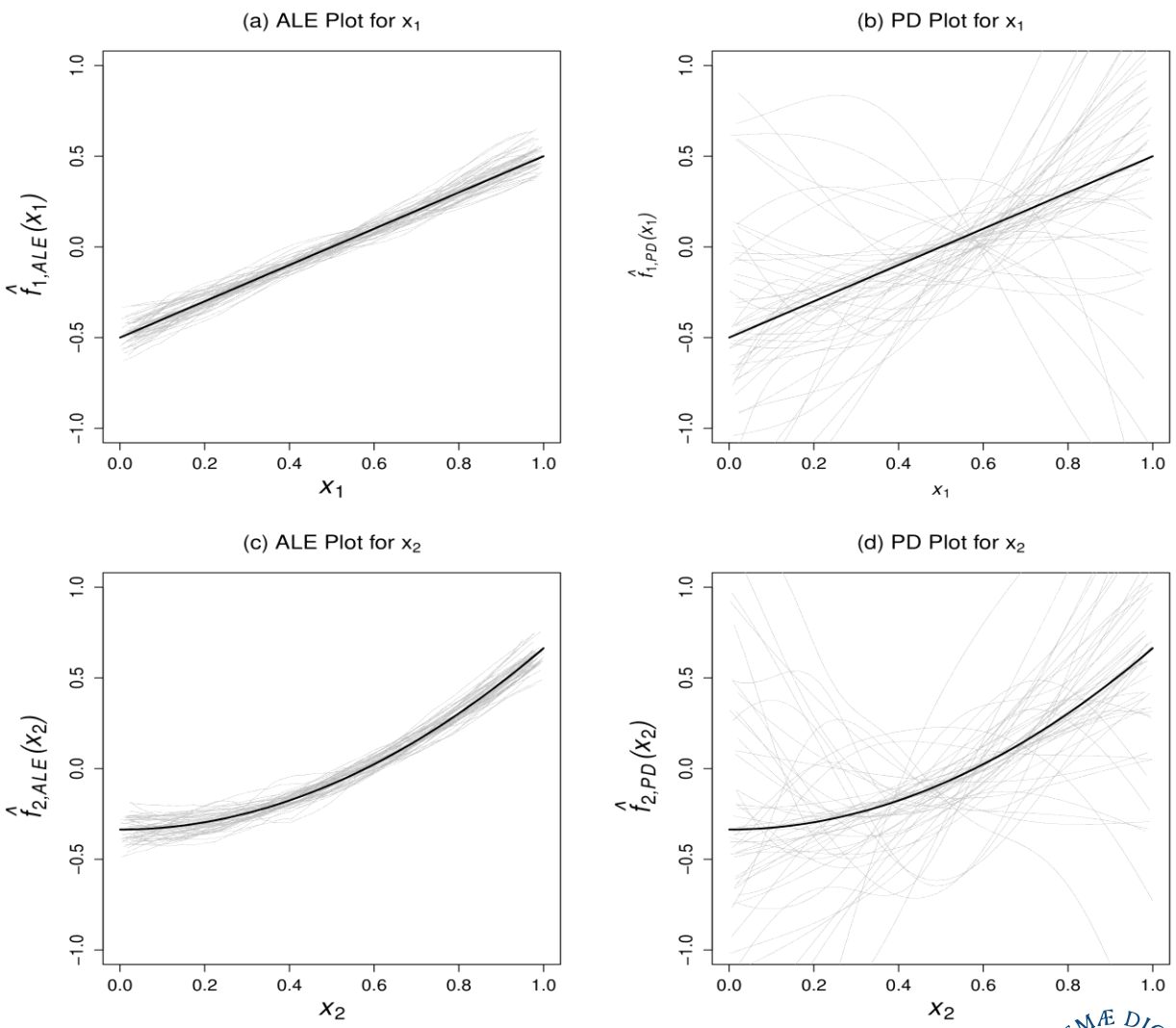
- Synthetic data generation:
  - d=2;
  - n=200;
  - (X_1, X_2) follows a uniform distribution along a segment of the line $x_1 = x_2$ with independent $N(0, 0.05^2)$ variables added to both predictors;
  - $Y = X_1 + X_2^2$
- Fit a tree using the **tree** package of R
  - overgrown and pruned back to 100 leaf nodes according to a cross-validation error sum of squares criterion
- For small x_1 values PD plot overestimates
- For small x_2 values PD plot underestimates
- M plots are quite close to each others because of the strong correlation between X_1 and X_2, and are each combinations of the true effects of X_1 and X_2.
- ALE and PD plotted with **ALEPlot** R package.



--- $\hat{f}_{j,M}(x_j)$
--- $\hat{f}_{j,PD}(x_j)$
— $\hat{f}_{j,ALE}(x_j)$
— True main effect of X_j
K = 50

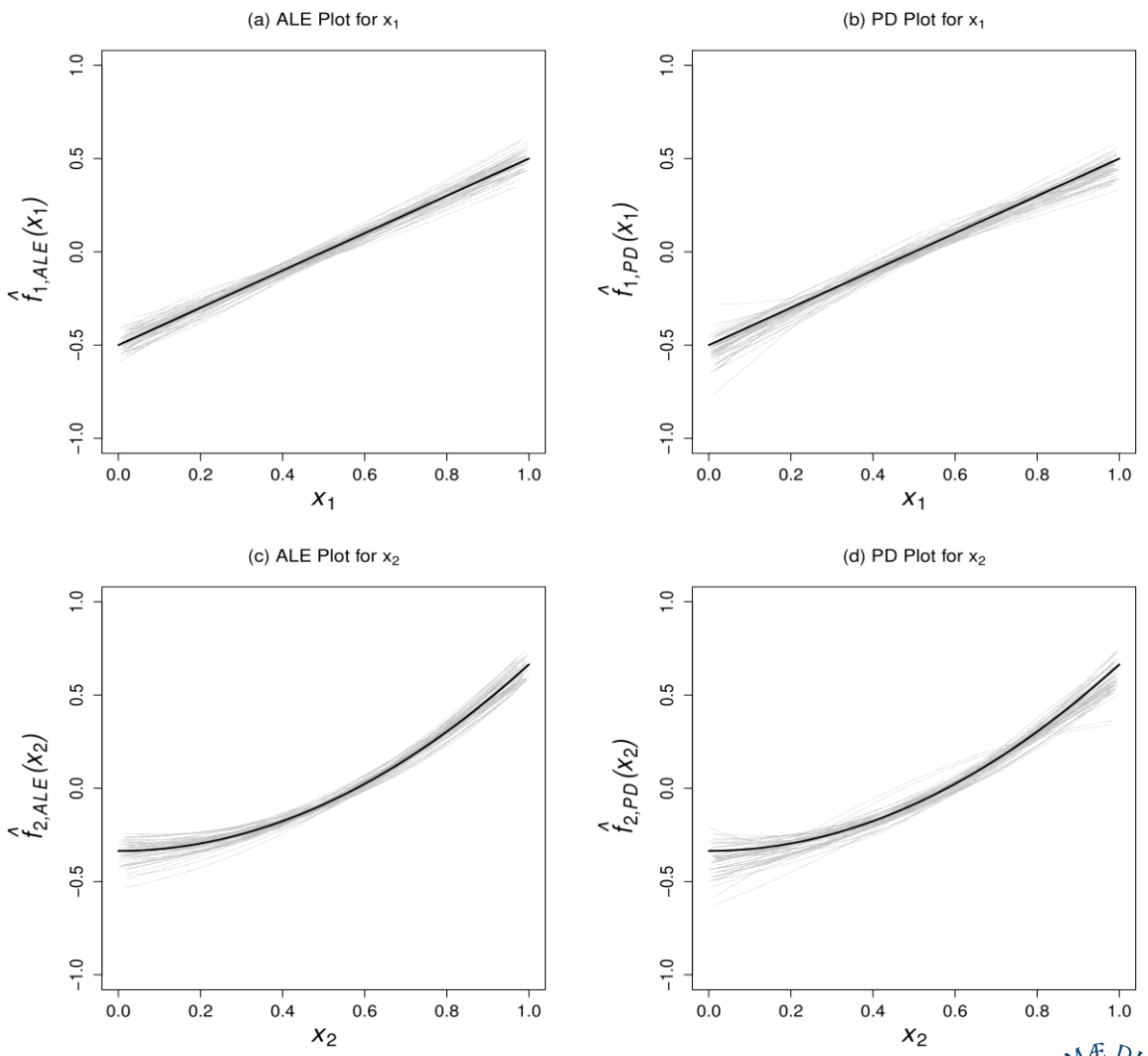
Experiment 2 – Neural Network model with synthetic data

- Synthetic data generation:
  - same as experiment 1
  - $Y = X_1 + X_2^2 + \varepsilon$ with $\varepsilon \sim N(0, 0.1^2)$
- Fit a neural network model with **nnet** package of R:
  - ten nodes in the single hidden layer;
  - decay parameter of 0.0001;
  - linear output activation function.
- Procedure repeated in a Monte Carlo simulation with 50 replicates, where on each replicate:
  - generate a new training data set of 200 observations;
  - refit the neural network with the same tuning parameters.
- On every replicate, $\hat{f}_{1,ALE}(x_1)$ and $\hat{f}_{2,ALE}(x_2)$ are quite close to the true linear and quadratic effects, respectively.
- $\hat{f}_{1,PD}(x_1)$ and $\hat{f}_{2,PD}(x_2)$ very inaccurate on many replicates.



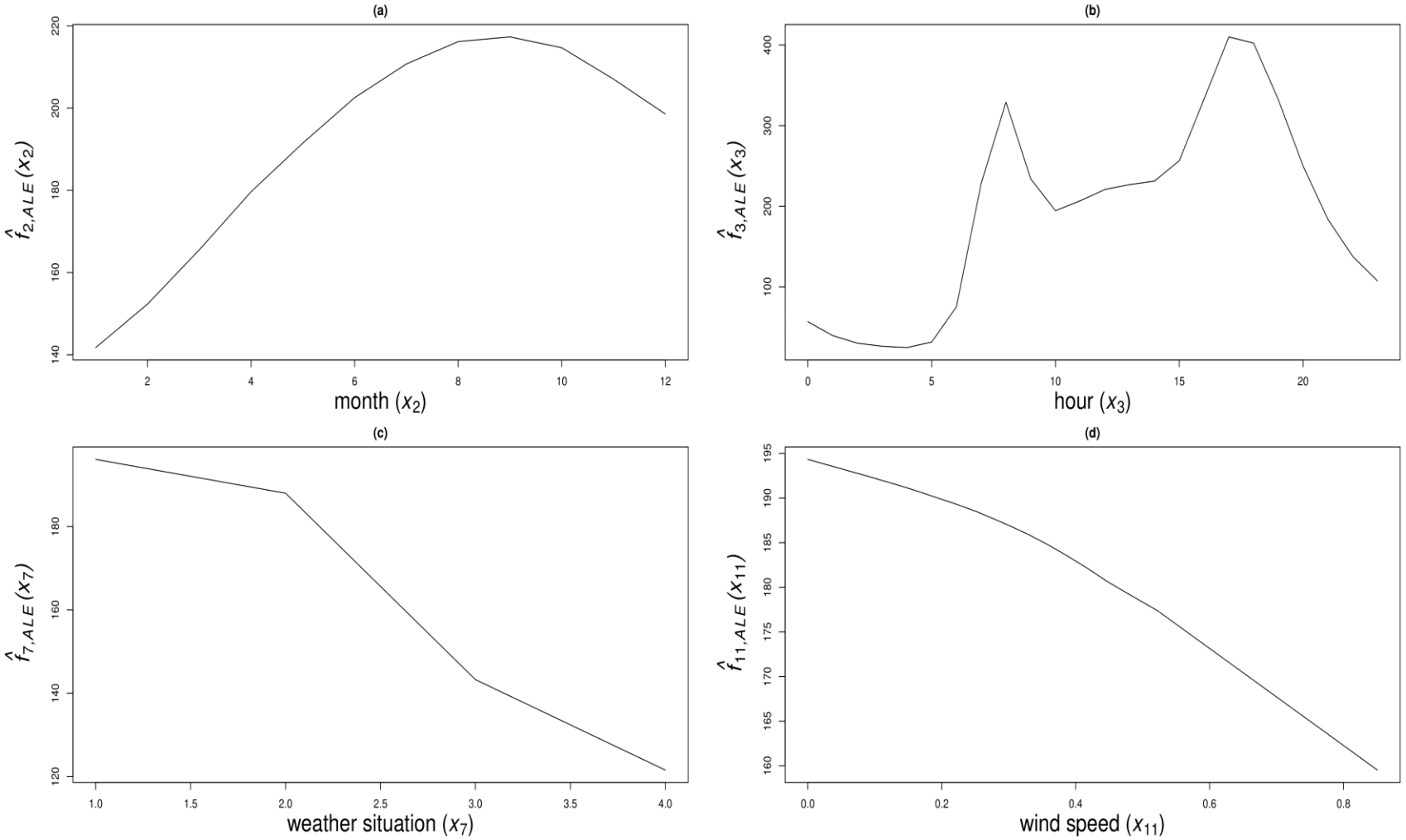
Experiment 2

- Experiment repeated with the same procedure.
- Hyperparameter tuning of the neural network with repeated 10-fold cross validation:
  - eight nodes in the single hidden layer;
  - decay parameter of 0.001;
  - linear output activation function.
- Theoretical r^2 value of $1 - \frac{\mathrm{Var}(\varepsilon)}{\mathrm{Var}(Y)} = 0.969$.
- Cross-validation r^2 range: $[0.960\ , 0.970]$.
- The discrepancy between ALE and PD plots is smaller, but still evident.
- PD plots interpretability strongly affected by the model stability.



Experiment 3 – a real data example

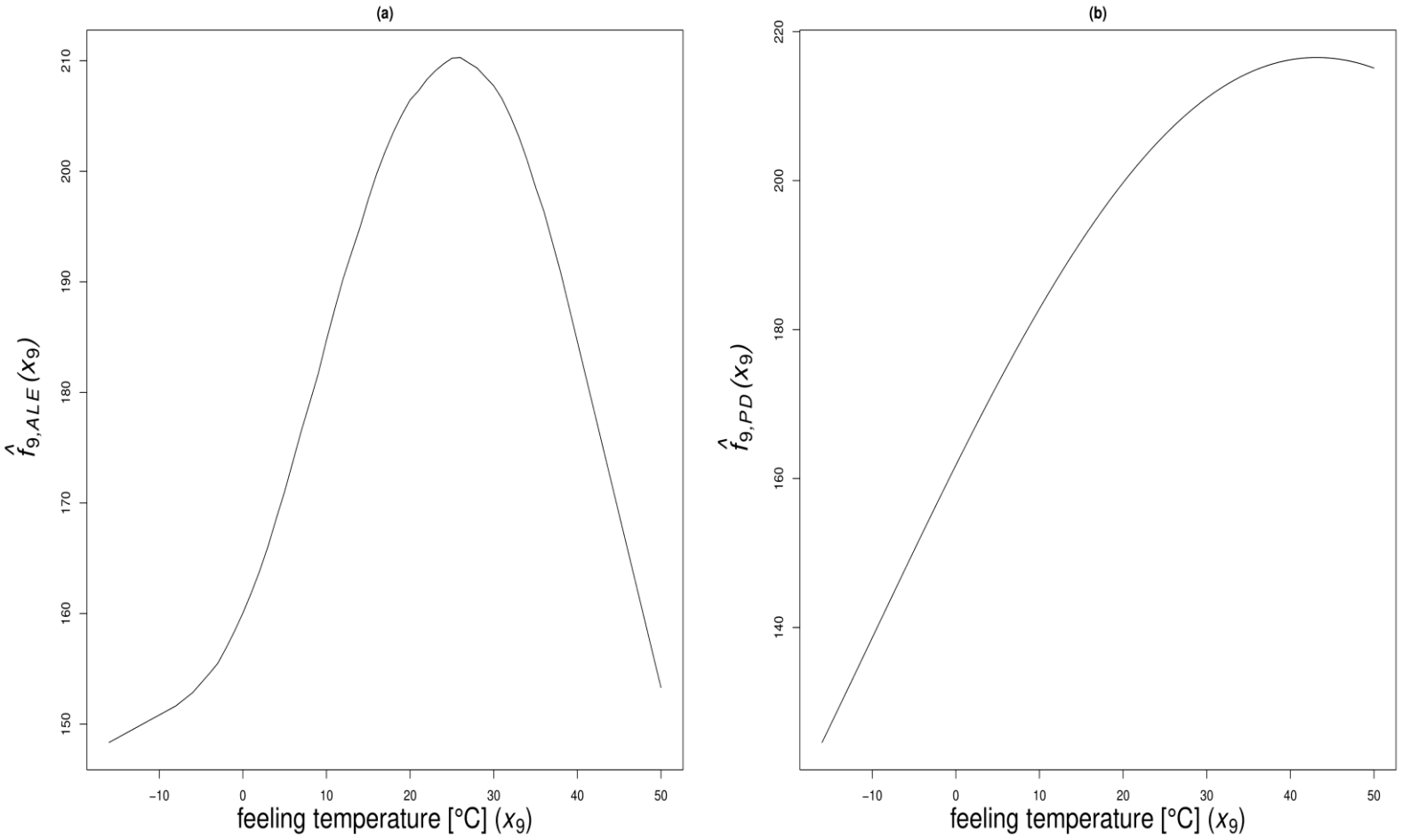
- Bike-sharing rental counts from the Capital Bikeshare system (Washington) in 2011-2012.
- The response is the total number of bike rental counts in each hour.
- d=11 predictors, n=17379.
- The set of predictors are highly correlated (e.g. $Corr(temp, atemp) \approx 0.99$).
- K = 100 for both main and interaction effect plots.
- Fit a neural network with:
  - ten nodes in the single hidden layer;
  - decay parameter of 0.05;
  - logistic output activation function.



- All of the shown ALE main-effect plots provide clear interpretations of the (main) effects of the predictors and are compatible with intuition.

Experiment 3 – ALE vs PD Plots

- For the effect of *atemp* (feeling temperature) the ALE plot is in much better agreement with common sense than the PD plot:
  - in the ALE plot the number of rentals steadily increases as *atemp* increases up until a comfortable temperature, after which it steadily decreases;
  - in the PD plot the number of rentals monotonically increases even at 40°C.
- The high correlation between *atemp* and some of the other predictors results in an extrapolation that makes PD plots unreliable.

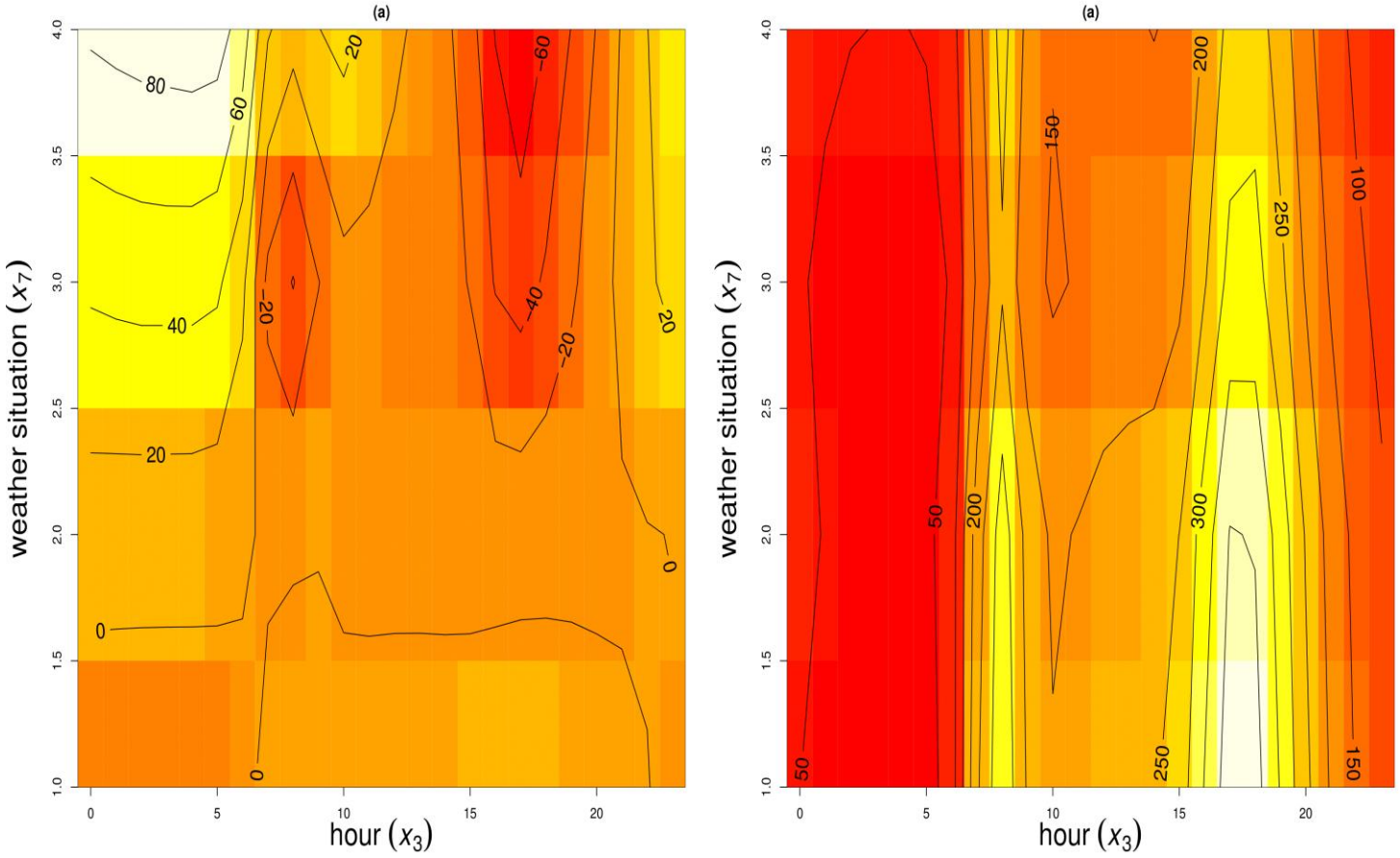


- The ALE plots are much faster to compute than PD plots:
  - ALE plot for *atemp*: 0.12 *s*
  - PD plot for *atemp*: 3.71 *s*

Experiment 3 – Second-order interaction plots

- ALE second-order interaction plots for the predictors hour and weather situation:
  - left panel: without the main effect of the two predictors included;
  - right panel: with the main effect of the two predictors included.
- They reveal a clear strong interaction between the two predictors:
  - the effect of weather situation is amplified at hours when the overall rental counts are expected to be higher.
- ATTENTION!

Without the main effects included, at some hours the function increases as weather situation increases (worsens), but when the main effects are included it is clear that increasing weather situation decreases bike rentals at any hour.



- Time to compute the interaction effect plots:
  - ALE plot: 0.22 *s*
  - PD plot: 330.55 *s*

Conclusions

- For visualizing the effects of the predictor variables in black box supervised learning models, PD plots are the most widely used method.
- ALE plots avoid the extrapolation problem that can render PD plots unreliable when the predictors are highly correlated.
- ALE plots avoid the omitted nuisance variable bias that render the M plots of little use for assessing the main effects of the predictors.
- ALE plots are much less computationally expensive than PD plots:
  - ALE plots require $2^{|J|} \times n$ evaluations of the supervised learning model $f(x)$ to compute each $\hat{f}_{J,ALE}(\mathrm{x}_J) \Rightarrow$ the number of evaluations does not depend on *K;*
  - PD plots require $K^{|J|} \times n$ evaluations to compute each $\hat{f}_{J,PD}(\mathrm{x}_J)$.

Reference: Daniel W. Apley, Jingyu Zhu, Visualizing the Effects of Predictor Variables in Black Box Supervised Learning Models, *Journal of the Royal Statistical Society Series B: Statistical Methodology*, Volume 82, Issue 4, September 2020, Pages 1059–1086, https://doi.org/10.1111/rssb.12377